ESSENTIALS OF ARTIFICIAL INTELLIGENCE

INDIVIDUAL CONTRIBUTION SHEET

Student Name	**D R Warnasooriya**	**Registration No.**	**D/BIT/24/0062**
Degree / Intake	**41 BIT**	**Group No.**	**32**
Project Title	**SmartStudy AI: A Hybrid Artificial Intelligence Academic Advisory System**		

Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Heuristic Safety System	Engineered deterministic time-math validation logic to block physically impossible schedules and trigger burnout/emergency alerts.	Rule-Based Reasoning (Expert System)	scripts/guardrails.py, UI
Output Sanitization	Developed a 10-step custom filtering pipeline to intercept raw AI outputs, stripping unescaped HTML and fixing formatting errors.	Regular Expressions (Regex)	sanitize_llm_output() function in app.py
PDF Rendering Engine	Programmed the automated document generator to transform the sanitized AI text into a highly formatted, downloadable multi-page report.	fpdf2, Custom StudyPlanPDF class	Generated PDF study plan files

End-to-End Pipeline Integration	Connected the raw LLM text generation stream to the regex sanitization module and PDF engine, ensuring seamless data flow from inference to export.	Python, Software Architecture Design	app.py integration logic,

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature:  **Date:** 17/10/2026

Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.